

FAIRFIELD COUNTY AP, OH, USA

WMO: 724294

Lat: 39.756N

Lon: 82.657W

Elev: 869

StdP: 14.24

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 53844

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.3	8.6	-5.6	4.2	3.8	0.4	5.8	9.7	23.8	36.9	20.7	37.1	2.9	260	0.471

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.6	90.3	74.1	88.2	73.3	85.8	72.0	77.1	86.7	75.5	84.0	74.2	82.1	8.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.1	131.2	82.0	72.7	125.4	80.4	71.8	121.5	79.3	40.9	86.4	39.5	84.5	38.2	81.9	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.0	17.7	15.8	-7.2	93.2	8.3	2.9	-13.2	95.3	-18.1	97.0	-22.8	98.6	-28.8	100.7
			-8.1	79.3	8.2	1.6	-14.0	80.4	-18.9	81.4	-23.5	82.2	-29.5	83.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.6	29.2	32.3	41.2	52.7	62.9	70.6	73.2	71.6	65.4	53.8	42.8	34.2
	DBStd	18.01	12.98	11.82	10.94	9.05	8.07	5.98	5.15	5.07	7.17	9.02	9.46	10.60
	HDD50	2374	652	505	318	74	5	0	0	0	1	61	256	502
	HDD65	5378	1109	916	739	377	140	20	3	7	81	364	666	956
	CDD50	3324	9	10	46	156	404	617	720	671	462	179	39	11
	CDD65	852	0	0	1	9	74	187	259	213	92	17	0	0
	CDH74	8141	0	1	21	122	743	1725	2354	1984	1009	179	3	0
	CDH80	2589	0	0	2	11	192	563	820	648	317	36	0	0
Wind	WSAvg	6.3	7.9	7.7	7.9	7.8	6.5	5.6	4.4	4.0	4.4	5.7	6.8	7.2
Precipitation	PrecAvg	40.8	3.0	2.2	3.4	3.9	4.2	4.1	4.2	3.5	3.4	2.9	2.7	3.1
	PrecMax	53.5	8.0	4.3	7.6	7.9	8.3	8.4	9.4	7.6	6.4	6.1	5.2	7.5
	PrecMin	30.2	1.2	1.0	1.6	1.0	1.0	0.9	2.3	1.3	1.2	0.9	0.8	1.3
	PrecStd	5.4	1.5	0.9	1.3	1.4	2.0	1.7	1.6	1.4	1.4	1.2	1.1	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.4	68.2	77.4	81.7	88.3	91.3	93.0	92.5	91.3	84.3	72.6	65.5
		MCWB	57.1	58.4	61.9	65.0	70.7	72.7	76.5	74.3	71.9	69.4	60.5	59.4
	2%	DB	59.3	61.3	70.8	77.9	84.5	88.5	90.0	89.5	87.6	79.0	67.4	59.9
		MCWB	54.6	53.4	58.1	61.5	68.5	72.7	75.8	74.0	71.4	66.0	57.3	54.0
	5%	DB	54.2	55.6	64.9	73.8	81.9	86.0	87.8	86.3	83.5	74.4	63.4	55.3
		MCWB	50.5	49.7	54.8	59.6	67.7	71.5	74.3	72.9	69.3	63.3	55.4	50.7
	10%	DB	48.0	50.0	60.4	70.0	78.7	83.3	84.6	83.6	80.4	70.1	59.1	50.5
		MCWB	43.4	44.2	52.6	58.0	66.5	70.4	72.8	71.5	67.4	61.4	52.4	46.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.1	59.9	63.4	68.3	73.9	77.0	79.8	78.3	75.9	71.9	63.8	60.1
		MCDB	63.2	66.8	75.6	78.6	84.1	86.9	88.6	87.7	85.8	81.7	68.8	64.6
	2%	WB	55.4	55.5	59.8	64.3	71.4	75.1	77.6	76.7	73.6	68.7	59.9	55.4
		MCDB	59.1	59.7	67.8	73.3	80.5	84.3	87.5	85.6	81.7	75.4	65.3	58.4
	5%	WB	50.7	50.2	56.9	61.9	69.4	73.5	76.0	75.0	71.8	65.5	56.5	51.3
		MCDB	53.6	55.3	63.4	69.8	78.4	82.8	84.6	82.7	79.8	72.0	62.1	54.7
	10%	WB	44.0	44.4	53.3	59.5	67.7	72.0	74.6	73.4	69.9	62.4	53.2	46.3
		MCDB	47.5	49.6	59.8	67.7	76.3	80.6	82.2	80.3	76.9	69.0	58.6	49.8
Mean Daily Temperature Range	5% DB	MDBR	16.2	17.6	19.6	23.2	21.9	20.7	20.6	22.1	24.2	22.8	19.8	15.3
		MCDBR	19.7	24.8	27.7	28.1	26.1	24.2	22.8	25.0	27.9	27.7	26.1	20.8
		MCWBR	16.5	19.1	17.3	15.6	12.7	10.9	10.4	11.7	13.2	15.8	17.9	17.2
	5% WB	MCDBR	18.2	23.1	23.2	23.0	21.5	19.8	19.7	19.9	21.9	23.0	21.3	18.8
		MCWBR	16.5	19.2	16.2	14.6	11.8	10.0	9.9	10.3	11.3	14.1	16.6	17.1
Clear-Sky Solar Irradiance	taub	0.309	0.326	0.355	0.392	0.436	0.456	0.466	0.458	0.399	0.365	0.326	0.304	
	taud	2.442	2.368	2.361	2.276	2.183	2.171	2.154	2.156	2.301	2.393	2.472	2.479	
	Ebn at Noon	272	282	285	280	269	263	259	257	266	265	264	266	
	Edh at Noon	26	31	36	41	46	47	47	46	37	30	25	23	
All-Sky Solar Radiation	RadAvg	524	779	1110	1500	1735	1901	1837	1717	1406	975	666	447	
	RadStd	51	106	101	134	144	129	102	97	130	89	89	51	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+284	N/A
(47)	Regional (36 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+234	+128

Nomenclature: See separate page